

Power analysis: optimisation of sea lice monitoring programme

Prevalence

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1 Preparation of sea lice dataset

R packages necessary of data analysis

```
library(ggplot2)
library(glmmTMB)
library(progress)
suppressMessages(library(dplyr))
suppressMessages(library(future.apply))
library(progressr)
```

1.1 Sea lice dataset

Sea lice monitoring dataset provided by the industry producer

```
Industry <- read.table(
  "Industry.csv",
  header = TRUE,
  sep = ',',
  fill = TRUE,
  dec = ".",
  na.strings = "NA"
)
```

2 Empirical data analysis

2.1 Modelling of temporal increase of sea lice prevalence levels

Adjustment of the variable pen: unique identification for each farm

```
Industry$Pen_un <- paste(Industry$SiteName, Industry$Pen.Count, sep = "_")
```

GLMM model abundance increase per production week, random effects to take into account the hierarchical structure of the data

```
mod_P <- glmmTMB(
  Prev_TF ~ Week_Production + (1 | SiteName) + (1 |
    Pen_un) + (1 | Year_Production),
  family = binomial,
  data = Industry
)
```

Exploration of the model

```
# Results overview
summary(mod_P)
```

```
## Family: binomial ( logit )
## Formula:
## Prev_TF ~ Week_Production + (1 | SiteName) + (1 | Pen_un) + (1 |
##   Year_Production)
## Data: Industry
##
```

```
##      AIC      BIC    logLik -2*log(L)  df.resid
##  86134.0  86183.1 -43062.0   86124.0   136122
##
## Random effects:
##
## Conditional model:
##   Groups      Name      Variance Std.Dev.
##   SiteName      (Intercept) 2.05619  1.4339
##   Pen_un        (Intercept) 0.05593  0.2365
##   Year_Production (Intercept) 1.53304  1.2382
## Number of obs: 136127, groups:  SiteName, 9; Pen_un, 96; Year_Production, 7
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -4.8003682  0.6854461   -7.0 2.5e-12 ***
## Week_Production 0.0645080  0.0006413  100.6 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# AIC
AIC(mod_P)
```

```
## [1] 86134
```

```
# Collinearity
performance::model_performance(mod_P)
```

```
## # Indices of model performance
##
## AIC      |      AICc |      BIC | R2 (cond.) | R2 (marg.) |  ICC |  RMSE
## -----|-----|-----|-----|-----|-----|-----
## 86133.998 | 86133.999 | 86183.105 | 0.621 | 0.202 | 0.526 | 0.314
##
## AIC      | Sigma | Log_loss | Score_log | Score_spherical
## -----|-----|-----|-----|-----
## 86133.998 | 1.000 | 0.315 | -Inf | 4.764e-05
```

2.2 Exploration of temporal increases in infection levels

2.2.1 Weekly increase for all farms

Odds ratio increase of prevalence converted to %

```
slope <- fixef(mod_P)$cond["Week_Production"]
OR <- exp(slope)
OR_increase <- (OR - 1) * 100
OR_increase
```

```
## Week_Production
##      6.663415
```

```

beta0 <- fixef(mod_P)$cond["(Intercept)"]
beta1 <- fixef(mod_P)$cond["Week_Production"]
p_base <- plogis(beta0)

logit_new <- beta0 + beta1
p_new <- plogis(logit_new)

perc_increase_prev <- ((p_new - p_base) / p_base) * 100
perc_increase_prev

```

```

## (Intercept)
##      6.605453

```

2.3 Weekly increase for each farm separately

GLMM of temporal increase for each farm

```

mod_P1 <- glmmTMB(
  Prev_TF ~ Week_Production + (Week_Production |
                                SiteName) + (1 | Pen_un) + (1 | Year_Production),
  family = binomial,
  data=Industry
)

```

Calculation of percentage increase, odds ratio increase of abundance converted to %

```

beta0 <- fixef(mod_P1)$cond["(Intercept)"]
beta1 <- fixef(mod_P1)$cond["Week_Production"]
ran <- ranef(mod_P1)$cond$SiteName

site_intercepts <- beta0 + ran[, "(Intercept)"]
site_slopes <- beta1 + ran[, "Week_Production"]

p_base <- plogis(site_intercepts)
p_new <- plogis(site_intercepts + site_slopes)

perc_increase_prev <- ((p_new - p_base) / p_base) * 100
abs_change_pp <- (p_new - p_base) * 100

# Results table
results <- data.frame(
  Site = rownames(ran),
  Baseline_Prevalence = p_base,
  New_Prevalence = p_new,
  Absolute_Change_pp = abs_change_pp,
  Percent_Increase_Prevalence = perc_increase_prev
)

results

```

```

##      Site Baseline_Prevalence New_Prevalence Absolute_Change_pp

```

## 1	a	0.027175185	0.028310948	0.11357634
## 2	b	0.010407178	0.011256276	0.08490980
## 3	c	0.004664479	0.004921613	0.02571335
## 4	d	0.006308824	0.006623095	0.03142709
## 5	e	0.008795638	0.009188732	0.03930933
## 6	f	0.101348276	0.109624050	0.82757742
## 7	g	0.003773982	0.004015482	0.02414998
## 8	h	0.031909438	0.033997456	0.20880179
## 9	i	0.014532372	0.014962726	0.04303541
##	Percent_Increase_Prevalence			
## 1		4.179414		
## 2		8.158773		
## 3		5.512587		
## 4		4.981450		
## 5		4.469185		
## 6		8.165678		
## 7		6.399071		
## 8		6.543575		
## 9		2.961348		

2.4 Prevalence variation between two consecutive inspections

Comparing prevalence levels between two consecutive visits in raw data

Prevalence at farm level

```
Ind1 <- aggregate(
  Industry$Prev_TF,
  list(
    Industry$SiteName,
    Industry$Year_Production,
    Industry$Week_Production,
    Industry$Count.Date
  ),
  mean
)
colnames(Ind1) <- c('Site', 'Year', 'Week', 'Date', 'Prev')

Ind1$Date <- as.Date(Ind1$Date, format = "%Y-%m-%d")
Ind1_s <- Ind1[order(Ind1$Date), ]
```

Calculation of percentage change in abundance levels

```
Ind1_s <- Ind1_s %>%
  arrange(Site, Date) %>%
  group_by(Site) %>%
  mutate(Prev_lag = lag(Prev),
    Increase = ifelse(Prev_lag == 0 | is.na(Prev_lag),
      NA, ((Prev - Prev_lag) / Prev_lag) * 100)
  ) %>% ungroup()

Ind1_s$Increase <- round(Ind1_s$Increase)
```

Exploration of prevalence change

```
Ind1_s1 <- as.data.frame(Ind1_s)
```

```
# Increase
```

```
summary(Ind1_s1[Ind1_s1$Increase>0,]$Increase)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##      1.0    25.0    63.5   144.2   148.5   2455.0    479
```

```
# Decrease
```

```
summary(Ind1_s1[Ind1_s1$Increase<0,]$Increase)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
## -100.00 -100.00  -50.00  -54.72  -22.75   -1.00    479
```

Increase in prevalence from 0

```
Ind_0<-Ind1_s[Ind1_s$Prev_lag == 0, ]
```

```
Ind_0 <- as.data.frame(Ind_0)
```

```
Ind_0<-Ind_0[!(is.na(Ind_0$Year)),]
```

```
nrow(Ind_0[Ind_0$Prev>0.09,])/nrow(Ind_0)
```

```
## [1] 0.02978723
```

2.5 Estimated parameters from GLMM

Slope: rate of prevalence increase

```
slope <- fixef(mod_P)$cond["Week_Production"]
```

Mean0: intercept on the logit scale

```
intercept_logit <- fixef(mod_P)$cond["(Intercept)"]
```

```
mean0 <- plogis(intercept_logit)
```

Random effects sd

```
site_sd <- sqrt(as.numeric(VarCorr(mod_P)$cond[["SiteName"]][1]))
```

```
pen_sd <- sqrt(as.numeric(VarCorr(mod_P)$cond[["Pen_un"]][1]))
```

```
year_sd <- sqrt(as.numeric(VarCorr(mod_P)$cond[["Year_Production"]][1]))
```

Number of fish per pen

```
fish_per_pen <- Industry %>%
  group_by(SiteName, Date, Pen.ID) %>%
  summarise(n_fish = n_distinct(FishID), .groups = "drop") %>%
  summarise(mean_fish = mean(n_fish)) %>%
  pull(mean_fish) %>%
  round()
```

Number of sites and years

```
n_sites <- length(unique(Industry$SiteName))
n_years <- length(unique(Industry$Year_Production))
```

3 Simulation-based power analysis on national level

3.1 Power analysis function

Function for simulation-based power analysis

```
simulate_test_s <- function(
  site_sd,
  pen_sd,
  year_sd,
  n_sites,
  n_pens,
  n_years,
  fish_per_pen,
  n_visits,
  mean0,
  slope,
  n_reps=100) {
  intercept <- qlogis(mean0)
  # Convert baseline prevalence from probability to logit scale

  # Run simulation
  power_results <- replicate(n_reps, { # run the simulation n_reps times

    # random effect
    site_re <- rnorm(n_sites, 0, site_sd)
    pen_re <- rnorm(n_sites * n_pens, 0, pen_sd)
    year_re <- rnorm(n_years, 0, year_sd)

    # study design
    dat <- expand.grid(
      Site = factor(1:n_sites),
      Pen = factor(1:n_pens),
      Week = 1:n_visits,
      Year = factor(1:n_years))

    # random effects
    dat$Pen_un <- interaction(dat$Site, dat$Pen)
    pen_levels <- interaction(dat$Site, dat$Pen)

    dat$site_re <- site_re[dat$Site]
    dat$pen_re <- pen_re[as.numeric(pen_levels)]
    dat$year_re <- year_re[dat$Year]

    # linear predictor
    eta <- intercept + slope * dat$Week +
      dat$site_re + dat$pen_re + dat$year_re
```

```

    dat$p <- plogis(eta)    # convert to probability

# Data generation
dat$infected <- rbinom(nrow(dat), size = fish_per_pen, prob = dat$p)
# simulate number of infected fish per week
dat$uninfected <- fish_per_pen - dat$infected
# Number of non-infected fish

# Model fitting
formula <- "cbind(infected, uninfected) ~ Week"

# Add random effects only if more than 1 level exists
if(n_sites > 1) formula <- paste0(formula, " + (1|Site)")
if(n_pens > 1) formula <- paste0(formula, " + (1|Pen_un)")
if(n_years > 1) formula <- paste0(formula, " + (1|Year)")

m <- try(glmmTMB(as.formula(formula), family = binomial, data = dat), silent = TRUE)

if (inherits(m, "try-error")) return(NA)    # If model fails to converge, skip this replica
if (!m$sdr$pdHess) return(NA) # skip non-converged models

# Detection rule
est <- summary(m)$coefficients$cond["Week", ]
# Extract estimate and p-value for week
est["Estimate"] > 0 && est["Pr(>|z|)"] < 0.05
})

# Remove NAs (failed fits)
power_clean <- power_results[!is.na(power_results)]

# Calculate power and confidence interval
power_estimate <- mean(power_clean)
ci <- binom.test(sum(power_clean), length(power_clean))$conf.int

list(power = power_estimate,
      lower_CI = ci[1],
      upper_CI = ci[2],
      n_reps = length(power_clean)
    )
}

```

3.2 Sampling scenarios

Odds ratios (effect size)

```
odds_ratios <- c(1.04, 1.06, 1.1, 1.2, 1.3)
```

Number of fish per pen

```
fish_vec <- c(5, 10, 15, 20, 25, 30)
```

Number of visits


```
visit_vec <- c(2, 3, 4, 6, 8)
```

Number of pens

```
pen_vec <- c(2, 5, 8, 10, 12, 15)
```

Replicates

```
n_reps <- 1000
```

3.3 Simulation run

In the following text the code to run simulation is provided, however, results are uploaded separately.

Create design grid for all scenarios

```
design_grid <- expand.grid(  
  OR = odds_ratios,  
  fish_per_week = fish_vec,  
  n_visits = visit_vec,  
  n_pens = pen_vec  
)
```

Prepare columns for results

```
design_grid$power <- NA_real_  
design_grid$lower_CI <- NA_real_  
design_grid$upper_CI <- NA_real_  
design_grid$n_success <- NA_integer_
```

Set parallel backend

```
# plan(multisession)
```

Reproducibility

```
# set.seed(123)
```

Progress bar

```
# handlers(global = TRUE)  
# handlers("txtprogressbar")
```

Simulation

```
#with_progress({  
  
# p <- progressr::progressor(steps = nrow(design_grid))
```

```

# results_list <- future_lapply(seq_len(nrow(design_grid)), function(i) {

  # Convert OR to slope
#   slope_i <- log(design_grid$OR[i])

  # Run simulation
#   sim_res <- simulate_test_s(
#     site_sd = site_sd,
#     pen_sd = pen_sd,
#     n_sites = n_sites,
#     year_sd = year_sd,
#     n_years = n_years,
#     n_pens = design_grid$n_pens[i],
#     fish_per_pen = design_grid$fish_per_week[i],
#     n_visits = design_grid$n_visits[i],
#     mean0 = mean0,
#     slope = slope_i,
#     n_reps = n_reps
#   )

#   p() # update progress

  # Return results
#   list(
#     power = sim_res$power,
#     lower_CI = sim_res$lower_CI,
#     upper_CI = sim_res$upper_CI,
#     n_success = sim_res$n_reps
#   )

# }, future.seed = TRUE)
#})

```

Adding results to prepared dataset

```

# design_grid$power      <- sapply(results_list, `[[`, "power")
# design_grid$lower_CI   <- sapply(results_list, `[[`, "lower_CI")
# design_grid$upper_CI   <- sapply(results_list, `[[`, "upper_CI")
# design_grid$n_success  <- sapply(results_list, `[[`, "n_success")

```

Calculate increase [%] from OR

```

# design_grid$perc_increase <- round((design_grid$OR - 1) * 100, 1)

```

3.3.1 Results of simulation

Loading saved results of simulation

```

design_grid <- readRDS("Prevalence_National.rds")

```

3.4 Results visualisation

Substitute 1 for 0.999

```
design_grid1 <- design_grid
if (any(design_grid1$power == 1, na.rm = TRUE)) {
  design_grid1$power[design_grid1$power == 1] <- 0.999
}
```

Final plot for abundance on national level

```
ggplot(design_grid1, aes(x = fish_per_week, y = n_pens, z = power)) +
  geom_contour_filled(breaks = seq(0, 1, by = 0.1), alpha = 1) +
  geom_contour(breaks = 0.8,
              color = "black",
              linewidth = 1.1) +
  facet_grid(
    perc_increase ~ n_visits,
    labeller = labeller(
      n_visits = function(x)
        paste(x, "visits"),
      perc_increase = function(x)
        paste0(x, "% increase")
    )
  ) +
  scale_fill_viridis_d(name = "Power", option = "A", begin = 0.15, end = 0.9) +
  labs(
    x = "Number of fish sampled per pen",
    y = "Number of pens"
  ) +
  theme_minimal(base_size = 10) +
  theme(panel.grid = element_blank(), strip.text = element_text(face = "bold"))
```

Minimal sample size

```
best_scenarios <- design_grid %>%
  mutate(No_Fish = fish_per_pen * n_visits * n_pens) %>%
  filter(power >= 0.8) %>%
  group_by(perc_increase, n_visits) %>%
  arrange(No_Fish, n_visits, n_pens, .by_group = TRUE) %>%
  slice(1) %>%
  ungroup() %>%
  arrange(perc_increase, n_visits, n_pens, No_Fish) %>%
  as.data.frame()

Best<-best_scenarios[,c(9,3,4,2,10,5)]
Best[,6]<-paste0(round(best_scenarios$power,2), ' [', round(best_scenarios$lower_CI,2), ', ',
               round(best_scenarios$upper_CI,2), ' ]')

knitr::kable(Best)
```

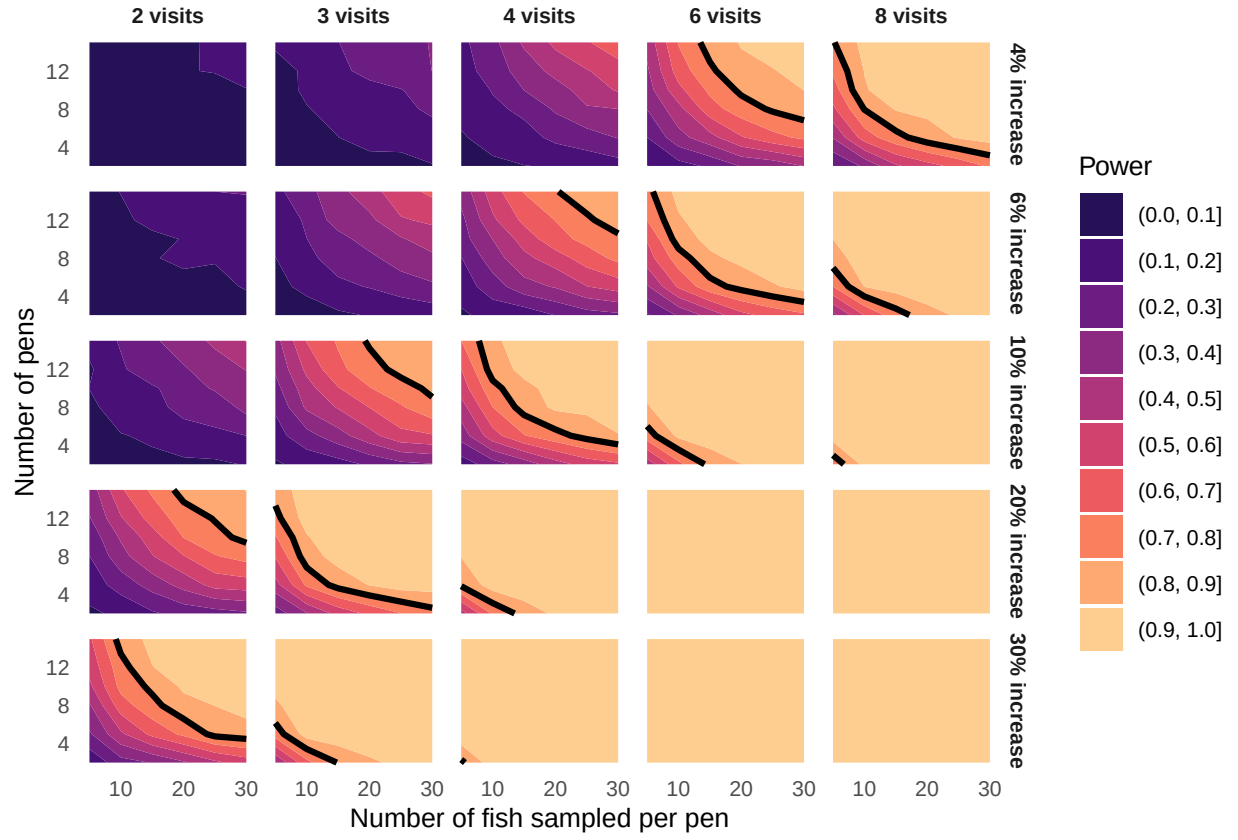


Figure 1: Power analysis for prevalence on national level

Table 1: Minimal sample size for prevalence at national level

perc_increase	n_visits	n_pens	fish_per_week	No_Fish	power
4	6	8	25	960	0.81 [0.79, 0.84]
4	8	5	20	800	0.85 [0.83, 0.87]
6	4	12	30	960	0.84 [0.82, 0.86]
6	6	5	20	600	0.83 [0.81, 0.85]
6	8	2	20	320	0.85 [0.83, 0.88]
10	3	10	30	600	0.82 [0.8, 0.85]
10	4	5	25	400	0.83 [0.81, 0.86]
10	6	2	15	240	0.83 [0.8, 0.85]
10	8	2	10	320	0.93 [0.91, 0.94]
20	2	10	30	400	0.82 [0.8, 0.85]
20	3	5	15	300	0.84 [0.81, 0.86]
20	4	2	15	160	0.84 [0.81, 0.86]
20	6	2	5	240	0.92 [0.9, 0.93]
20	8	2	5	320	0.99 [0.99, 1]
30	2	5	25	200	0.83 [0.8, 0.85]
30	3	2	15	120	0.81 [0.78, 0.83]
30	4	2	10	160	0.95 [0.94, 0.97]
30	6	2	5	240	0.99 [0.99, 1]
30	8	2	5	320	1 [1, 1]

4 Simulation based-power analysis on farm level

4.1 Sampling scenarios

Farm level: 1 farm per 1 production year

```
n_sites <- 1  
n_years <- 1
```

Odds ratio (Effect size)

```
odds_ratios <- c(1.2, 1.3, 1.5, 2, 2.5)
```

Number of fish

```
fish_vec <- c(5, 10, 15, 20, 25, 30)
```

Number of visits

```
visit_vec <- c(4, 6, 8, 10, 12)
```

Number of pens

```
pen_vec <- c(2, 5, 8, 10, 12, 15)
```

Replicates

```
n_reps <- 1000
```

4.2 Simulation run

In the following text the code to run simulation is provided, however, results are uploaded separately.

Create design grid for all scenarios

```
design_grid <- expand.grid(  
  OR = odds_ratios,  
  fish_per_week = fish_vec,  
  n_visits = visit_vec,  
  n_pens = pen_vec  
)
```

Prepare columns for results

```
design_grid$power <- NA_real_  
design_grid$lower_CI <- NA_real_  
design_grid$upper_CI <- NA_real_  
design_grid$n_success <- NA_integer_
```

Set parallel backend

```
# plan(multisession)
```

Reproducibility

```
# set.seed(123)
```

Progress bar

```
# handlers(global = TRUE)
# handlers("txtprogressbar")
```

Simulation

```
#with_progress({

# p <- progressr::progressor(steps = nrow(design_grid))

# results_list <- future_lapply(seq_len(nrow(design_grid)), #function(i) {

  # Convert OR to slope
#   slope_i <- log(design_grid$OR[i])

  # Run simulation
#   sim_res <- simulate_test_s(
#     site_sd = site_sd,
#     pen_sd = pen_sd,
#     n_sites = n_sites,
#     year_sd = year_sd,
#     n_years = n_years,
#     n_pens = design_grid$n_pens[i],
#     fish_per_pen = design_grid$fish_per_week[i],
#     n_visits = design_grid$n_visits[i],
#     mean0 = mean0,
#     slope = slope_i,
#     n_reps = n_reps
#   )

#   p() # update progress

  # Return results
#   list(
#     power = sim_res$power,
#     lower_CI = sim_res$lower_CI,
#     upper_CI = sim_res$upper_CI,
#     n_success = sim_res$n_reps
#   )

# }, future.seed = TRUE)

#})
```

Adding results to a prepared dataset

```
# design_grid$power      <- sapply(results_list, `[[`, "power")
# design_grid$lower_CI   <- sapply(results_list, `[[`, "lower_CI")
# design_grid$upper_CI   <- sapply(results_list, `[[`, "upper_CI")
# design_grid$n_success  <- sapply(results_list, `[[`, "n_success")
```

Calculate percent increase from OR for clarity

```
# design_grid$perc_increase <- round((design_grid$OR - 1) * 100, 1)
```

4.2.1 Results of simulation

Loading saved results of simulation

```
design_grid <- readRDS("Prevalence_Farm.rds")
```

4.3 Results visualisation

Substitute 1 for 0.999 due to visualization

```
design_grid1 <- design_grid
if (any(design_grid1$power == 1, na.rm = TRUE)) {
  design_grid1$power[design_grid1$power == 1] <- 0.999
}
```

Visualisation of results

```
ggplot(design_grid1, aes(x = fish_per_week, y = n_pens, z = power)) +
  geom_contour_filled(breaks = seq(0, 1, by = 0.1), alpha = 1) +
  geom_contour(breaks = 0.8,
               color = "black",
               linewidth = 1.1) +
  facet_grid(
    perc_increase ~ n_visits,
    labeller = labeller(
      n_visits = function(x)
        paste(x, "visits"),
      perc_increase = function(x)
        paste0(x, "% increase")
    )
  ) +
  scale_fill_viridis_d(name = "Power", option = "A", begin = 0.15, end = 0.9) +
  labs(
    x = "Number of fish sampled per pen",
    y = "Number of pens"
  ) +
  theme_minimal(base_size = 10) +
  theme(panel.grid = element_blank(), strip.text = element_text(face = "bold"))
```

Calculation of minimal sample size

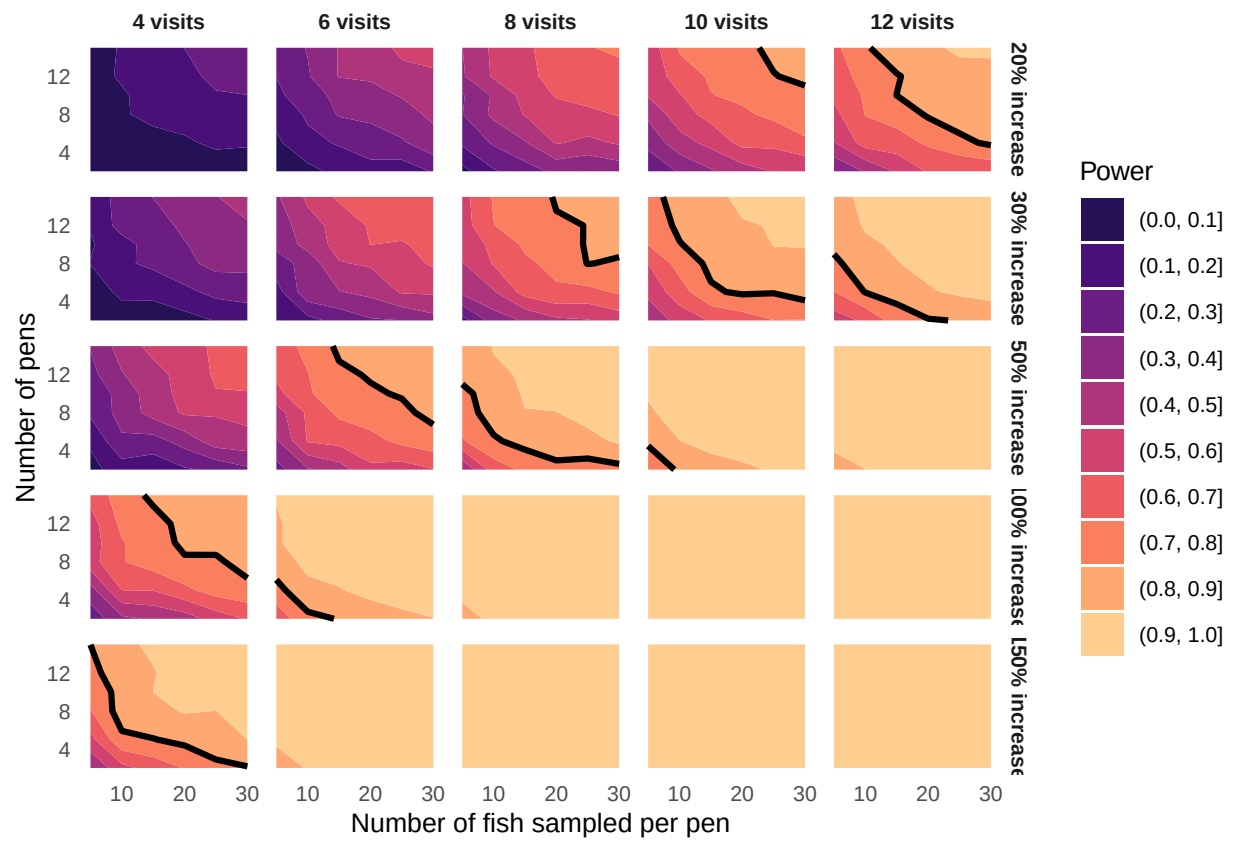


Figure 2: Power analysis for prevalence at farm level

Table 2: Minimal sample size for prevalence at farm level

perc_increase	n_visits	n_pens	fish_per_week	No_Fish	power
20	10	12	30	2400	0.83 [0.8, 0.85]
20	12	5	30	1200	0.82 [0.79, 0.84]
30	8	8	25	1280	0.8 [0.78, 0.83]
30	10	5	20	1000	0.81 [0.79, 0.84]
30	12	2	25	480	0.8 [0.78, 0.83]
50	6	8	30	960	0.83 [0.8, 0.85]
50	8	5	15	800	0.83 [0.81, 0.86]
50	10	2	10	400	0.82 [0.79, 0.84]
50	12	2	5	480	0.85 [0.82, 0.87]
100	4	8	30	640	0.84 [0.82, 0.86]
100	6	2	15	240	0.8 [0.78, 0.83]
100	8	2	5	320	0.83 [0.81, 0.86]
100	10	2	5	400	0.96 [0.95, 0.98]
100	12	2	5	480	0.99 [0.98, 1]
150	4	5	20	400	0.83 [0.8, 0.85]
150	6	2	5	240	0.81 [0.79, 0.84]
150	8	2	5	320	0.97 [0.95, 0.98]
150	10	2	5	400	0.99 [0.98, 1]
150	12	2	5	480	0.99 [0.98, 1]

```

best_scenarios <- design_grid %>%
  mutate(No_Fish = fish_per_pen * n_visits * n_pens) %>%
  filter(power >= 0.8) %>%
  group_by(perc_increase, n_visits) %>%
  arrange(No_Fish, n_visits, n_pens, .by_group = TRUE) %>%
  slice(1) %>%
  ungroup() %>%
  arrange(perc_increase, n_visits, n_pens, No_Fish) %>%
  as.data.frame()

Best<-best_scenarios[,c(9,3,4,2,10,5)]
Best[,6]<-paste0(round(best_scenarios$power,2), ' ', round(best_scenarios$lower_CI,2), ', ',
  round(best_scenarios$upper_CI,2), ' ')

```

```
knitr::kable(Best)
```